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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	**HTL**
Selected Domain	**Domain 2 - ALPHA RED**
Selected SDG Track (1–17)	**SDG 8 – Decent Work and Economic Growth**
Project / Solution Name	**Quora Chat**
Name (Leader)	**Tng Kar Ming**
Email & Phone (Leader)	**karming0425@gmail.com / +60173553022**
University Affiliation(s)	**Asia Pacific University of Technology & Innovation**
Number of Team Members	**5**

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro, small and medium enterprises (MSMEs) play a significant role in Malaysia's economy, representing 96.1% of business establishments in 2024 (OECD, 2026) and contributing 39.7% of Malaysia's GDP in 2025 (DOSM, 2026). However, smaller businesses often operate with limited manpower while managing customer enquiries alongside daily operations.

One employee may handle multiple customer conversations simultaneously, including enquiries in English, Bahasa Melayu and Chinese. Each enquiry requires requirement clarification, quotation preparation, managerial approval and customer follow-up. When information is scattered across lengthy conversations, employees may overlook requirements, prepare inaccurate quotations or delay responses. Unsigned quotations can also be forgotten, potentially resulting in missed sales opportunities.

These problems extend beyond the quotation stage. Once a customer accepts a quotation, employees must communicate the agreed requirements to the operational team. Without a consistent handover process, important commitments may be overlooked, causing repeated clarification, misunderstandings and additional administrative work.

Although customer relationship management and automation platforms exist, implementing multiple disconnected systems can introduce unnecessary complexity for resource-constrained SMEs.

The core challenge is therefore enabling small teams to handle increasing customer enquiries while preserving accurate requirements, maintaining human control over commercial decisions and reducing repetitive administrative work.

Addressing this challenge supports SDG 8, particularly Targets 8.2 and 8.3, by improving productivity through technological innovation and supporting the sustainable growth of micro and small businesses.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Quora Chat is an AI-assisted customer enquiry and quotation automation module designed for the Orbyte SME ecosystem. Rather than functioning as a standalone chatbot, it connects customer communication, business knowledge, quotation approval and operational handover through one traceable workflow.

The solution consists of four interconnected modules:

1. Knowledge Base: Businesses maintain public product information and internal operating procedures. AI assistants retrieve relevant knowledge according to their permitted context, supporting consistent responses while separating customer-facing information from internal guidance.

2. Quotation Inbox: A multilingual AI assistant collects customer requirements through natural conversation, asks clarifying questions and prepares structured requirement briefs. Employees can review extracted information, identify missing details, monitor customer activity and intervene when human judgment is required.

3. Quotes: Confirmed requirements are converted into quotation drafts using deterministic pricing rules or staff-provided estimates. Authorized employees review, approve, reject or revise quotations before customers receive PDF documents and eSign links. Automated reminders support follow-up on unsigned quotations.

4. Team Chat: After customer acceptance, the system creates an operational job and prepares an AI-assisted handover containing the agreed requirements, scope and customer commitments. Employees can collaborate internally, consult relevant business knowledge and request customer clarification when necessary.

The workflow also supports invoices, simulated payments and receipts, demonstrating continuity beyond quotation acceptance.

Using generative AI APIs, PostgreSQL, deterministic business logic and Replit hosting, Quora Chat demonstrates an automated journey from customer enquiry to an approved outcome and recorded handover. The solution aims to reduce active employee handling time per quotation by 30% while preserving requirement completeness, approval control and traceability. This is a proposed evaluation target rather than an achieved result.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Component	Technology	Purpose
Frontend	TanStack Start, React, TypeScript	Customer and company-facing interfaces
Styling/UI	Tailwind CSS, shadcn/ui	Responsive and accessible user interfaces
API	Hono, oRPC	HTTP services and type-safe communication
Database	PostgreSQL, Drizzle ORM	Persistent customer, quotation, workflow, and audit records
Generative AI	Replit-managed OpenAI / OpenRouter	Multilingual conversations, requirement extraction, summarization, and drafting
Business Logic	TypeScript	Deterministic pricing, validation, approval rules, and workflow transitions
Background Processing	PostgreSQL-backed worker	Handover processing, retries, and asynchronous workflow tasks
Document Generation	React PDF	Quotation, agreement, and billing PDF generation
Hosting	Replit	Deployment of the functional application
Version Control	GitHub	Source code management and collaboration

Sponsor Platform Integration: Replit hosts the functional prototype and provides managed AI integration. The application uses PostgreSQL for persistent workflow data.

System Architecture: The customer-facing interface gathers requirements using public business knowledge. Structured information enters the Orbyte Quotation Inbox for employee review before quotation preparation, approval and customer eSign. Following acceptance, the workflow creates an operational job and transfers confirmed requirements into Team Chat.

Implementation Boundaries: WhatsApp, Orbyte services, notifications and payment processing are simulated using synthetic data. The prototype does not require access to the production Orbyte ecosystem.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team followed an iterative design-thinking approach focused on reducing repetitive customer quotation work for resource-constrained SMEs.

Empathize and Define: We examined the customer enquiry-to-quotation process and identified recurring challenges involving incomplete requirements, multilingual communication, manual approvals, delayed follow-ups and fragmented operational handovers.

Ideate: Rather than building a standalone chatbot, we designed a reusable Orbyte module connecting customers, AI assistants, employees and managers through a shared workflow.

Prototype: We developed two interfaces and four interconnected modules: Knowledge Base, Quotation Inbox, Quotes and Team Chat. Five synthetic SME environments demonstrate how the solution can adapt to different business categories.

Test: As of 24 October 2026, the project passed 63 unit tests and 36 database integration tests covering requirement consistency, quotation approvals, concurrent messages, signing, workflow recovery and knowledge access boundaries. These results validate tested application behavior, not real-world AI accuracy.

Deployment: The functional prototype is hosted on Replit with PostgreSQL and managed AI integration. Key remaining risks include AI extraction errors, external API availability and simulated identity management. Mitigations include structured validation, deterministic pricing, human approval, retry mechanisms and replay-based demonstration.

Phase 2 will prioritize usability evaluation, measurable productivity improvements and further AI reliability testing.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8.2:** Improve economic productivity through technological upgrading and innovation. **SDG 8.3:** Support productive activities, entrepreneurship and the growth of micro and small enterprises. Quora Chat addresses these targets by reducing repetitive quotation administration and improving SME workflow efficiency.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Estimated 20 minutes of active employee time per completed quotation** in a manual SME workflow. This includes customer requirement clarification, quotation preparation, approval coordination and operational handover. The figure is a team-defined planning estimate for a representative synthetic scenario, not a measured industry average.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Average active employee handling time per completed quotation (minutes).** The target is a reduction of 6 minutes, equivalent to 30%. Supporting KPIs are requirement completeness of at least 95% and 100% compliance with mandatory human quotation approvals during controlled prototype tests.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Reduce active staff time from an estimated 20 minutes to 14 minutes per completed quotation, representing a 30% improvement.** Maintain at least 95% completeness of mandatory customer requirements and zero unauthorized quotation approvals in controlled synthetic scenarios. These are proposed evaluation targets, not achieved results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code framework	MIT
TanStack Start / Router	Code framework	MIT
Hono	Code framework	MIT
oRPC	Code framework	MIT
Drizzle ORM	Code framework	Apache License 2.0
PostgreSQL	Database	PostgreSQL License
Tailwind CSS	UI framework	MIT
shadcn/ui and underlying components	UI code/components	Verify applicable component licenses
React PDF	PDF generation library	MIT
OpenAI GPT-5 mini	Pre-trained AI model/API	Proprietary API/service terms
Replit	Cloud development and hosting platform	Commercial service terms
OpenRouter	AI API platform, local development	Platform service terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Tng Kar Ming

Tng

24 October 2026

8. Appendix

8.1 Measurement Methodology

The team will compare manual and Quora Chat-assisted workflows using identical synthetic customer enquiries. Each scenario will cover requirement clarification, quotation preparation, approval and operational handover.

Active employee time will exclude periods spent waiting for customer responses or automated background execution. Requirement completeness will be calculated as the percentage of mandatory fields resolved before quotation approval. Approval compliance will measure whether consequential quotation decisions follow the required human review process.

The primary KPI will be calculated using:

Time Reduction (%) = (Manual Staff Time − AI-Assisted Staff Time) ÷ Manual Staff Time × 100

Results will be recorded across repeated scenarios using consistent assumptions. The initial 20-minute baseline and 30% target will be updated when measured evidence becomes available.

8.2 System Architecture

